

Wisdom

What the BN Flow Knows

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Abstract

The Beurling–Nyman flow is not just a reformulation of the Riemann Hypothesis. It is a process that *knows* things: it knows where the primes are, it knows the zeros of ζ , it knows the arithmetic of division. We identify precisely what the flow knows at each step, what it learns by adding $\rho_{1/N}$, and what it would need to know to prove RH. The wisdom of the flow: it encodes the Euler product at every step, it separates the arithmetic (primes, factorization) from the analytic (Mellin, ζ), and it makes RH a statement about whether its knowledge is ever complete. The wall is not ignorance. The wall is the limit of what arithmetic can see in ζ .

1 What the Flow Knows

Definition 1 (Knowledge at step N). At step N , the BN flow has:

- *The span* $\mathcal{V}_N = \text{span}\{\rho_{1/2}, \dots, \rho_{1/(N+1)}\}$: the subspace generated by the first N BN functions.
- *The residual* $r_N = 1_{(0,1)} - f_N^* \perp \mathcal{V}_N$: the part of $1_{(0,1)}$ not yet known.
- *The distance* $d_N = \|r_N\|$: how much remains unknown.
- *The alignment angles* $\cos^2 \theta_1, \dots, \cos^2 \theta_N$: how much each step learned.

The flow’s knowledge of $1_{(0,1)}$ at step N is the projection f_N^* . The flow’s ignorance at step N is the residual r_N .

Proposition 2 (What $\rho_{1/k}$ knows about k). *The BN function $\rho_{1/k}(x) = \{1/(kx)\} - (1/k)\{1/x\}$ encodes:*

1. *The divisibility structure of k : the function $x \mapsto \{1/(kx)\}$ has period- k structure in the variable $1/x$, tracking multiples of $1/k$.*
2. *The arithmetic of k relative to 1: the subtraction $(1/k)\{1/x\}$ normalizes so that $\rho_{1/k}$ is orthogonal to the constant function.*

3. *The Dirichlet series of k : via the Mellin transform, $\widehat{\rho}_{1/k}(s) = (k^{-s} - k^{-1})\zeta(s)/s$, connecting k 's position in the integers to the full spectrum of ζ .*

At each step, the flow absorbs the arithmetic of one more integer k . By step N , it has absorbed $\{2, 3, \dots, N+1\}$.

*Remark 3 (What composites teach). When $k = ab$ is composite, $\rho_{1/k}$ carries information about the interaction of a and b — how they divide the same set of multiples. Numerically: this interaction is already (approximately) known from $\rho_{1/a}$ and $\rho_{1/b}$. The composite adds the *correlation* between a and b , not the independent information. The flow discards the redundant part (via Gram-Schmidt) and retains only the new correlation.*

2 What Primes Teach

Proposition 4 (Primes teach irreducible arithmetic). *For prime p , $\rho_{1/p}$ carries information about p that is not deducible from $\rho_{1/2}, \dots, \rho_{1/(p-1)}$: namely, the p -th prime factor of $1/\zeta$.*

Via the Euler product: $1/\zeta(s) = (1 - 2^{-s})(1 - 3^{-s})(1 - 5^{-s}) \dots$. Each prime p contributes the factor $(1 - p^{-s})$. The flow “learns” $(1 - p^{-s})$ when it adds $\rho_{1/p}$. Before adding $\rho_{1/p}$, the flow approximates $1/\zeta$ as if p did not exist. After adding $\rho_{1/p}$, it knows about the p -th prime.

This is why primes give large alignment angles: $\cos^2 \theta_{p-1}$ measures how much the p -th prime matters — how different $1/\zeta$ is from its approximation without the p -th factor.

Theorem 5 (The flow knows PNT after the first prime). *After adding $\rho_{1/2}$, the flow knows: $1/\zeta(s) \approx (1 - 2^{-s})$ on the critical line. The error is $d_1^2 = 1 - \log 2 \approx 0.307$: ignoring all primes but 2, the flow is 30% wrong.*

After adding all primes up to p : $1/\zeta(s) \approx \prod_{q \leq p} (1 - q^{-s})$. The error $d_{\pi(p)}^2$ measures how much the primes above p contribute.

As $p \rightarrow \infty$: $\prod_{q \leq p} (1 - q^{-s}) \rightarrow 1/\zeta(s)$ (in the half-plane $\text{Re}(s) > 1$). On the critical line ($\text{Re}(s) = 1/2$): the convergence is the content of RH.

The prime number theorem (ζ has no zeros on $\text{Re}(s) = 1$) tells the flow: the product of $(1 - p^{-s})$ for $p \leq P$ is a reasonable approximation to $1/\zeta(s)$ for $\text{Re}(s)$ bounded away from 1. RH says: even on $\text{Re}(s) = 1/2$.

3 What Zeros Know

Proposition 6 (The zeros as gaps in knowledge). *Each zero $\rho_k = 1/2 + i\gamma_k$ of ζ corresponds to a “frequency” γ_k at which the flow is wrong: $\zeta(1/2 + i\gamma_k) = 0$, so $1/\zeta$ has a pole at γ_k , and the BN optimal polynomial $P_N^*(1/2 + i\gamma_k)$ can be anything (since $\zeta P_N^* = 0$ at γ_k regardless of P_N^*).*

The function $x^{-\rho_k} 1_{(0,1)}(x)$ is in $\mathcal{V}_\infty^\perp$ if and only if ρ_k is an off-line zero ([4]). This means: off-line zeros create permanent blind spots — frequencies at which the flow cannot accumulate knowledge.

Under RH ($\text{Re}(\rho_k) = 1/2$): the zeros are on the critical line. The function $x^{-1/2 - i\gamma_k} 1_{(0,1)}$ is NOT in $\mathcal{V}_\infty^\perp$ (since $\langle x^{-1/2 - i\gamma_k}, \rho_{1/n} \rangle \neq 0$ in general). The flow has no permanent blind spots. It can accumulate knowledge at all frequencies γ_k .

Under $\neg RH$: there is a permanent blind spot. The residual cannot go below the size of the blind-spot component. The flow knows everything except the off-line zero.

Remark 7 (The wisdom of the zeros). The zeros on the critical line do not block the flow; they slow it. At each zero γ_k , the flow has reduced gain: $|P_N^*(1/2 + i\gamma_k)|$ is bounded but does not help (since $\zeta(1/2 + i\gamma_k) = 0$). The flow must learn about the neighborhood of γ_k , not the zero itself. As the zeros become denser (at height T , density $\sim \log T/2\pi$), learning their neighborhoods requires more steps. But the density is only logarithmic — the flow can keep up.

The wisdom of the zeros on the critical line: their density is logarithmic, not polynomial. There is just barely enough room for the flow to learn all of $1/\zeta$. Move one zero off the critical line, and the balance is broken.

4 What RH Says

Theorem 8 (RH as completeness of knowledge). *The following are equivalent:*

1. *The Riemann Hypothesis.*
2. $d_N \rightarrow 0$: *the flow eventually knows all of $1_{(0,1)}$.*
3. $\sum \cos^2 \theta_N = \infty$: *the flow never stops learning.*
4. $\mathcal{V}_\infty = \overline{\mathcal{V}_\infty}$: *the span of all BN functions is dense in the part of $L^2(0, \infty)$ that matters.*
5. *The flow has no permanent blind spots (no off-line zeros).*
6. *The arithmetic of the integers is complete: every frequency γ_k that ζ knows about is accessible from the multiplicative structure.*

RH says: the primes, by generating all integers, generate all the analytic information in ζ on the critical line. Arithmetic (multiplication, divisibility) is sufficient. Analysis (zeros, poles) contains nothing beyond the arithmetic.

Remark 9 (The Dirichlet interpretation). The Dirichlet series $\zeta(s) = \sum_n n^{-s}$ encodes all positive integers. The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ encodes only primes. They are equal. This equality IS the unique factorization theorem.

RH says: on the critical line, this equality is “tight” in a spectral sense. The spectrum of the integers (via ζ) is exactly what the spectrum of the primes (via the Euler product) implies. No zero can “hide” off the critical line, because off-line zeros would mean the integers contain analytic information not derivable from the primes alone.

In the BN flow: the flow builds the Euler product step by step. When it completes, it must match $1/\zeta$ exactly. RH is the statement that this completion is faithful: the product of prime factors, built one prime at a time, converges to the true reciprocal.

5 What We Know

Theorem 10 (The wisdom of the approach). *The BN approach has established:*

1. *The flow is analytic: d_N decreases monotonically; each step is a projection; the residual is orthogonal to all previous steps.*
2. *The flow knows arithmetic: primes give large angles; composites give small corrections; the Euler product structure is visible.*
3. *The continuum model is complete: for the exact Toeplitz operator with symbol σ , the prediction error is 0 unconditionally (from $\int \log \sigma = -\infty$, Widom–KWK).*
4. *The discretization is the wall: the BN Gram matrix differs from the continuum model by the arithmetic of ζ 's zeros. This correction is small under RH (polynomially in $\log N$), and order 1 under $\neg$ RH.*
5. *Numerical evidence is strong: $d_N \rightarrow 0$ computationally; $S_N \sim \log N / \log \log N$; prime structure dominates.*

What remains:

1. *Prove that $d_N^2 \leq C(\log N)^{2-\delta}$ for some $\delta > 0$, unconditionally. This would give $\sum \cos^2 \theta_N \geq \sum_p (\log p)^\delta / p = \infty$.*
2. *Prove the lower bound on the oscillatory integral $\int H_{N-1}(t) e^{-it \log N} dt$ without assuming RH.*
3. *Transfer the Killip–Simon divergence from the exact Toeplitz model to the BN matrix.*

Each of these is equivalent to RH — there is no shortcut. The wall is real.

Remark 11 (The shape of the wall). The wall has been seen from many angles:

- From the Gram matrix: the condition number of G_N grows as $N^2 \log N$ (under RH) vs. faster (under $\neg$ RH).
- From the Toeplitz model: the non-Toeplitz correction is small (under RH) vs. order 1 (under $\neg$ RH).
- From the Verblunsky coefficients: the sum $\sum |\alpha_N^{\text{BN}}|^2$ diverges (under RH) vs. converges (under $\neg$ RH).
- From the oscillatory integral: the sum $\sum_k e^{-i\gamma_k \log N}$ has square-root cancellation (under GUE, consistent with RH) vs. does not (under $\neg$ RH, where one zero has wrong sign).
- From the prime structure: the prime contributions dominate, and their sum grows as $\sim \log N / \log \log N$ (diverging).

In every picture, the wall is the same: whether the arithmetic of the integers (primes, factorization) is sufficient to determine the analytic structure of ζ on the critical line. The BN flow makes this question precise. The answer is: yes, if and only if RH.

6 The Limit of Knowledge

Remark 12 (What the flow cannot know). The BN flow, at each step, adds one more integer. It accumulates arithmetic knowledge. What it cannot know, ever, is whether a zero is off the critical line — until it has accumulated all integers (step $N = \infty$).

This is the fundamental asymmetry: $\neg$ RH is a global statement (one zero, somewhere, is off the line) while the BN flow is a local process (one integer at a time). The flow must detect a global pathology through local accumulation.

RH says: the accumulation is sufficient. The local steps, taken together, are globally faithful. The arithmetic contains no hidden structure that would let a zero escape to $\text{Re}(s) \neq 1/2$.

This is the deepest content of RH: *arithmetic is globally self-consistent on the critical line*.

Theorem 13 (The wisdom statement). *Let K_N be the arithmetic knowledge accumulated by the BN flow after N steps: the projection of $1_{(0,1)}$ onto $\mathcal{V}_N = \text{span}\{\rho_{1/2}, \dots, \rho_{1/(N+1)}\}$. Then:*

$$RH \iff \lim_{N \rightarrow \infty} K_N = 1_{(0,1)} \iff \text{arithmetic knowledge of the integers is complete.}$$

The wisdom: completeness of arithmetic knowledge is the Riemann Hypothesis. This is not a proof. It is a precise statement of what is being asked. The wall is where knowledge ends. What lies beyond the wall is RH.

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